An Explainable AI-Based Decision Support System for Dynamic Water Quality Index Prediction and Recommendation

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Abstract

Water quality is characterized by multiple physical and chemical parameters. This study presents an explainable AI-based decision-support system for estimating the constructed station-year Water Quality Index (WQI) from six predictors: BOD, DO, Ammonia-N, Nitrite-N, Nitrate-N, and pH. Five parameters, excluding DO, construct the target. The study uses Northern Ireland Environment Agency (NIEA) observations from 1990 to 2024; it estimates the index for the represented station-year rather than forecasting future water quality.

The work includes data preprocessing, missing-value analysis, WQI construction, machine-learning and deep-learning model comparison, model validation, explainable AI, and recommendation generation. Five ML models—Random Forest, Extra Trees, Gradient Boosting, XGBoost, and CatBoost—were compared with Deep Neural Networks and TabNet. These models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2.

The CatBoost model showed strong performance and was selected for the explainability and decision-support stages. Five-fold cross-validation was performed to assess the consistency of the CatBoost model. SHAP was used to understand how the input parameters contributed to individual predictions and to study the overall importance of the features. By combining the predicted WQI, parameter values, model explanations, and guideline exceedance information, a recommendation engine was developed.

The complete system was implemented using Streamlit. In the dashboard section, the user enters the water-quality parameters as input and receives the predicted WQI, water-quality category, guideline analysis, SHAP explanation, and recommendation. Therefore, the system connects water-quality prediction, explainability, and practical decision support in a single workflow.

Keywords: Water Quality Index, Water Quality Assessment, Water Quality Prediction, Machine Learning, Deep Learning, Explainable Artificial Intelligence, SHAP, Decision Support System

1 Introduction

Water is one of the most important natural resources for human life, ecological systems, agriculture, and industrial activities. Water quality directly influences aquatic ecosystems, public health, and the sustainable management of water resources. However, water quality can change substantially because of natural processes and

human activities. Agricultural runoff, wastewater discharge, industrial activities, urban development, and changes in environmental conditions can introduce different pollutants into rivers and other water bodies. At the same time, natural processes such as seasonal variation, hydrological changes, and geological characteristics can affect the measured properties of water. For these reasons, continuous and systematic water-quality monitoring is very important for understanding changes in water conditions and achieving effective environmental management. Water quality is generally analyzed using a combination of physical, chemical, and biological parameters. Individual parameters provide information about different characteristics of a water body. Biochemical oxygen demand (BOD) is associated with the amount of oxygen required for the biological decomposition of organic matter, while dissolved oxygen (DO) provides information about the oxygen conditions available in the aquatic environment. Nitrogen-related parameters such as ammonia-nitrogen, nitrite-nitrogen, and nitrate-nitrogen can provide information about nutrient loading and possible pollution. Similarly, pH represents the acidity or alkalinity of the water and can affect chemical reactions and aquatic organisms. Although each parameter provides useful information, interpreting a large number of measurements independently can be difficult, particularly when observations are collected from many monitoring locations over long periods. A large monitoring dataset contains thousands of observations with spatial and temporal variation, making it difficult to obtain a clear representation of the overall water-quality condition from individual measurements. Water Quality Index (WQI) methods have been developed to address this problem by combining information from multiple water-quality parameters into a single numerical index. A WQI provides a clear representation of overall water quality and can make water-quality information easier to communicate and interpret. Different WQI approaches have been proposed using different parameter sets, reference objectives, weighting methods, normalization procedures, and aggregation strategies. Consequently, the same monitoring data can produce different index values depending on the selected WQI formulation. It is important to clearly define how the index is constructed and what the resulting value represents. In this study, the Canadian Council of Ministers of the Environment Water Quality Index (CCME-WQI) framework is used to construct the station-year water-quality target. The CCME-WQI considers the scope, frequency, and amplitude of exceedances relative to defined objectives, providing a representation of water-quality conditions. The WQI used in this study is a constructed index rather than a directly measured variable, and the prediction task is designed to estimate this constructed station-year index. The availability of long-term environmental monitoring datasets has created opportunities to apply data-driven methods to water-quality assessment. Conventional assessment approaches generally require repeated examination of multiple parameters and become increasingly difficult as the volume and dimensionality of monitoring data increase. Machine-learning (ML) methods provide an alternative approach by learning relationships between input variables and a target variable from observed data. In water-quality applications, regression algorithms can be used to identify nonlinear relationships among chemical and physical parameters and an overall water-quality indicator. Previous studies have investigated different machine-learning techniques for water-quality prediction and WQI estimation, demonstrating that ensemble-learning methods can be useful for modelling complex relationships between water-quality variables. However, model performance depends on several factors, including the characteristics of the dataset, preprocessing procedures, selected predictors, target definition, and experimental design. Differences in these factors can make direct comparisons between studies difficult. Another important challenge in environmental machine learning is the interpretation of model predictions. A model may produce a highly accurate numerical prediction, but prediction accuracy alone does not explain why a particular result was obtained. This is important in water-quality assessment because users need to understand which parameters contribute to a poor or good predicted condition. For example, if a model predicts a low WQI, the numerical prediction itself does not indicate whether the prediction is primarily associated with nitrate, nitrite, ammonia, BOD, pH, or another input variable. Without an interpretable mechanism, a

machine-learning model can remain difficult to use in practical environmental decision-making. Explainable Artificial Intelligence (XAI) methods can help address this issue by providing information about the contributions of individual input features to model predictions. In this study, SHapley Additive exPlanations (SHAP) are used to interpret the selected prediction model and examine how individual water-quality parameters contribute to the predicted WQI values. The need for interpretation becomes even more important when machine-learning models are intended to support practical decisions rather than only produce numerical predictions. A predicted WQI can indicate the estimated quality category of a station-year condition, while SHAP can provide information about the parameters contributing to the prediction. However, prediction and explanation alone still do not provide sufficient information for a user who needs to determine what should be examined next. A decision-support system should connect the numerical prediction with parameter-level assessment and an understandable interpretation of the observed condition. This motivates an integrated workflow in which water-quality measurements are first processed to obtain the model inputs, the WQI is predicted, the prediction is explained using SHAP, potentially important parameters are assessed against the selected guideline objectives, and structured recommendations are generated. The workflow can be represented as: Water-quality measurements → WQI prediction → SHAP explanation → parameter assessment → recommendation Such an integrated approach aims to bridge the gap between predictive modelling and practical interpretation rather than treating prediction, explainability, and decision support as separate tasks. Previous research has demonstrated the value of machine learning for water-quality prediction and the usefulness of XAI for interpreting model behaviour, but differences in datasets, preprocessing strategies, target definitions, evaluation procedures, and validation designs remain important considerations. Even high predictive accuracy does not by itself establish geographic or temporal generalization. These considerations are particularly relevant when models are developed using environmental monitoring data collected from multiple locations and years. The study addresses these challenges through an integrated explainable artificial intelligence framework using long-term river-water-quality observations from the Northern Ireland Environment Agency (NIEA). The project dataset covers observations from 1990 to 2024 and includes measurements from a large number of monitoring stations. The long observation period and multiple monitoring locations provide a heterogeneous dataset containing variation across both stations and years. From the available water-quality measurements, BOD, DO, ammonia-nitrogen, nitrite-nitrogen, nitrate-nitrogen, and pH are used as predictors for the machine-learning and deep-learning models. Five of these parameters—BOD, ammonia-nitrogen, nitrite-nitrogen, nitrate-nitrogen, and pH—are used to construct the CCME-WQI target, while DO is included as a predictor in the modelling stage. The prediction task estimates the constructed station-year WQI from six predictor variables rather than forecasting a future water-quality condition. An important component of the proposed workflow is the treatment of incomplete monitoring observations. Long-term environmental datasets commonly contain missing measurements because monitoring frequency can vary among stations and parameters, and individual measurements may not be available for every sampling event. Ignoring missingness without assessment can reduce the number of usable observations and potentially affect WQI construction and predictive modelling. Therefore, the present study includes a missing-value analysis and applies parameter-specific imputation procedures to recover selected missing measurements. The resulting datasets are evaluated through separate baseline and imputed modelling branches, allowing the effect of the preprocessing strategy to be examined rather than assuming that a single preprocessing choice is universally optimal. This provides a more transparent basis for evaluating WQI construction and the predictive models. Following preprocessing and WQI construction, several machine-learning and deep-learning approaches are compared under a common experimental framework. The machine-learning models include Random Forest, Extra Trees, Gradient Boosting, XGBoost, and CatBoost, while Deep Neural Networks (DNN) and TabNet are included as deep-learning approaches. The models are evaluated using Mean Absolute

Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Using the same target formulation, predictor set, train-test split, and evaluation metrics provides a consistent basis for comparing different modelling approaches. The comparison aims to identify a suitable predictive model for the explainability and decision-support stages rather than assuming that an algorithm will always perform best across environmental datasets. Among the evaluated models, CatBoost is selected for the explainability and decision-support stages based on its strong predictive performance in the experimental comparison. Its performance is further examined using five-fold cross-validation to assess the consistency of the model across different subsets of the available station-year records. SHAP analysis is then applied to the selected CatBoost model to determine the contributions of the six predictor variables to individual WQI predictions. This provides both a global perspective on feature importance and a local perspective on how individual measurements influence specific predictions. These model-derived contributions are interpreted as predictive associations rather than direct evidence of environmental causation. This distinction is important because an explanatory model can indicate which variables influence the model output without proving that a particular variable is the physical cause of the observed water-quality condition. The final component of the proposed framework is a recommendation-oriented decision-support layer. The system combines the predicted WQI category, input parameter values, selected guideline-exceedance information, and model explanations to generate structured recommendations. The purpose of these recommendations is not to replace environmental investigation or regulatory decision-making, but to provide an interpretable assessment that can help users identify parameters requiring attention and consider appropriate monitoring or source-investigation actions. This distinction is important because a machine-learning prediction and a guideline exceedance do not by themselves establish the exact source of pollution or prescribe a scientifically validated remediation action. Finally, the recommendation component is designed as decision support rather than as an autonomous environmental management system. The complete workflow is implemented through an interactive Streamlit dashboard. The dashboard allows a user to enter the six water-quality predictor values and obtain a predicted WQI, corresponding water-quality category, parameter-level guideline assessment, SHAP-based explanation, and structured recommendation as output. By integrating these components into a single interface, the system moves beyond a conventional prediction-only application. Instead, it provides a sequence from measurement input to prediction, explanation, assessment, and recommendation. This design makes the output of the machine-learning model easier to understand for users who do not have direct access to the underlying model implementation. Overall, this study develops an explainable AI-based decision-support framework for station-year WQI estimation using long-term NIEA river-water-quality observations. The work combines data preprocessing and missing-value analysis, CCME-WQI construction, comparative machine-learning and deep-learning modelling, CatBoost validation, SHAP-based interpretation, parameter-level assessment, recommendation generation, and interactive dashboard deployment. The central contribution is not limited to obtaining a numerical WQI prediction. Instead, the study connects water-quality data, index estimation, predictive modelling, model interpretation, and practical decision support within a single overall workflow. By maintaining an explicit distinction between constructed WQI estimation, model-derived explanation, guideline assessment, and recommendation, the proposed framework provides a transparent basis for applying machine learning to water-quality assessment and monitoring.

2 Related Work / Literature Review

2.1 Traditional Water Quality Index Approaches

Over the years, many WQI methodologies have been developed, each addressing different parameter sets, reference standards, normalization techniques, weighting strategies, and aggregation procedures. Consequently, identical water-quality datasets may result in different index values based on the selected WQI formulation.

The WQI is a constructed composite indicator rather than a directly measured variable. Yan et al. [7] discuss uncertainties in WQI models and limitations of machine-learning applications. In this study, the chosen formulation and objectives define the constructed target that the regression models estimate.

The Canadian Council of Ministers of the Environment Water Quality Index (CCME-WQI) incorporates scope, frequency, and amplitude of objective exceedances [12]. This study applies the framework to Northern Ireland Environment Agency (NIEA) observations [11], using the parameter set and objectives configured in the project implementation [13]; their status as statutory limits has not been established.

2.2 Machine Learning for Water Quality Prediction

Machine-learning methods have been studied for WQI estimation. Choudhary et al. [8] combined six regression learners in a stacked ensemble and used SHAP to interpret model predictions. This verified example motivates comparing regression models while keeping the target formulation and explanation scope explicit.

Yan et al. [7] review machine-learning approaches for water-quality prediction, including indicator and index prediction, and discuss limitations concerning data processing, model uncertainty, and generalization. This review does not establish geographic transfer of the models developed in the present project.

There are several limitations:

- Model performance is highly dependent on the characteristics of the dataset
- Predictive accuracy may vary across different geographic locations
- Spatial and temporal generalization remains a critical concern
- High numerical performance does not necessarily equate to scientific validity
- Interpretability of the models is also essential.

2.3 Deep Learning for Water Quality Prediction

Deep-learning models have also gained attention in water-quality prediction because water-quality conditions are influenced by various environmental factors that interact in complex ways.

Li et al. [6], in a 2024 review published in Environmental Science and Pollution Research, examine deep-learning methods for water-quality prediction and forecasting, including single-model and mixed-model approaches, optimization, and data fusion. Their review discusses data scarcity and long-term prediction challenges; these broader forecasting applications are distinct from the station-year index estimation implemented here.

Performance depends on dataset size, feature characteristics, temporal and spatial structures, model architecture, preprocessing procedures, and hyperparameter configuration.

ML and DL should be experimentally compared under the same evaluation conditions.

2.4 Explainable Artificial Intelligence and SHAP

Machine-learning models provide accurate WQI predictions, but their outputs are often difficult to interpret. In water-quality applications, understanding the factors that influence model predictions is important, as these results can support environmental assessment and water-resource management.

Explainable AI (XAI) methods help describe model predictions. SHAP provides feature attributions to individual predictions [1], and Choudhary et al. [8] apply it to WQI regression. Pandey et al. [9] identify explainable AI as a direction for further water-quality prediction research. These attributions explain model outputs, not physical causality.

2.5 Decision-Support Systems

Previous studies have shown that ML and DL models can be used to predict water quality, while XAI methods can provide insight into the parameters influencing these predictions. However, prediction and explanation alone do not form a complete decision-support framework. This provides the motivation for developing an integrated approach that combines prediction, explainability, and decision-oriented analysis.

The aim is to connect constructed station-year WQI estimates and model explanations within a decision-support workflow. Malisaba et al. [10] discuss methodological and operational gaps, including reliance on localized datasets and limited

assessment beyond accuracy metrics. These concerns motivate transparent reporting of the validation and advisory limitations of the present system.

3 Materials and Methods

Proposed Methodology

Figure 1 presents the project workflow linking constructed station-year WQI estimation, model explanation, and advisory outputs. The six-predictor regression task is index estimation, not future forecasting.

3.1 Dataset Description

Our study uses long-term river-water-quality monitoring data from the Northern Ireland Environment Agency (NIEA). The public source is catalogued as *River Water Quality Monitoring 1990 to 2024 – All Parameters* [11]. The project dataset covers 1990 to 2024 and contains observations from multiple river monitoring stations across Northern Ireland. The original project dataset has 178,680 observations and 40 variables, including station information, geographic information, sampling dates, monitoring conditions, water-quality measurements, and associated quality indicators.

The dataset contains observations from 1,311 monitoring stations, with both spatial and temporal variation in water-quality conditions.

3.2 Data Preprocessing

3.2.1 Data Inspection

An initial inspection was performed to understand the structure and quality of the raw observations. The data types, missing values, parameter ranges, station identifiers, and temporal coverage were examined before further preprocessing.

The raw dataset contains measurements collected over more than three decades. The sampling dates range from January 1990 to December 2024, while the monitoring records are distributed across 1,311 stations.

The inspection also identified differences in parameter availability. Some parameters have a large proportion of missing observations, whereas others are available for most monitoring records. Therefore, parameter selection was performed before constructing the WQI target.

3.2.2 Data Cleaning and Transformation

The raw observations were processed to obtain a suitable dataset for WQI construction. Relevant columns were retained, while unnecessary fields were removed from the modelling dataset. Date information was converted into a usable temporal representation, and the year of observation was extracted for station-year analysis.

The original dataset contains 178,680 observation rows and 40 columns. The baseline preprocessing script converts Date using `pd.to_datetime(errors="coerce")` and extracts Year. It retains 13 columns: StationCode, Location, PrimaryBasin, GridReference, Date, Time, Year, and the six measurements BOD, DO, Ammonia-N, Nitrite-N, Nitrate-N, and pH. The only explicit row filter is `dropna` with `how="all"` on these six measurements. Exactly 3,065 rows have all six measurements missing, leaving 175,615 rows in the saved core dataset. Rows with only some measurements missing are retained. The script does not delete rows because of extreme values, quality flags, duplicates, or individual missing parameters.

The final imputation script instead reads the original 178,680 rows, derives Year and Month, and retains all rows in its saved output. It does not apply the baseline all-six-missing filter. The baseline WQI builder therefore receives 175,615 core observation rows, whereas the imputed WQI builder receives 178,680 observation rows. This difference is a branch-specific preprocessing choice, not a change in the number of observations in the original source file. The imputation flags identify 10,178 recovered parameter measurements; this is a count of filled measurement cells, not additional observation rows.

Missing values were not blindly replaced with zero, since zero represents a valid measurement for some parameters and can introduce artificial observations. The missing measurements were first analyzed according to their temporal, station-wise, and cross-parameter patterns. To recover potentially useful information, two complementary imputation strategies were evaluated: parameter-based predictive imputation and station-based temporal imputation. The approaches were validated using MAE, RMSE, and R^2 before being applied to the final dataset.

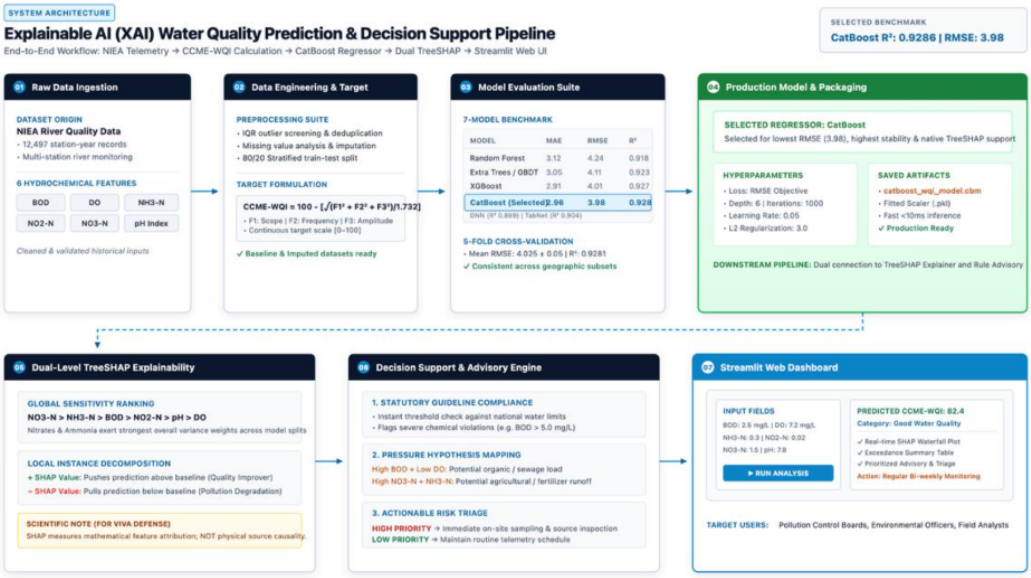


Fig. 1 System architecture of the proposed explainable AI-based water quality decision support system.

3.2.3 Quality Control

The supplied quality-control scripts inspect the data and produce reports; they do not write a filtered observation dataset. The WQI builders read the core or imputed dataset directly, without an additional quality-control row-removal stage.

The following checks are performed:

- Missing-value analysis
- Negative-value analysis
- Extreme-value detection
- Duplicate-record inspection
- Examination of original quality-flag fields

No negative values were found for the selected water-quality parameters during the validation stage.

The extreme-value script reports measurements above its inspection thresholds but does not remove them. The value-validation script reports negative-value and duplicate counts without dropping rows; the supplied core dataset has no negative values in the six selected measurements and no duplicate rows. These diagnostics do not account for any additional reduction from 175,615 baseline core observations.

The original NIEA quality-flag fields are inspected by reporting their frequencies among observed measurements. These include $<$ and $>$ qualifiers. No flag-based exclusion, qualifier-dependent adjustment, or censoring correction is implemented by the supplied WQI-construction scripts.

3.3 Missing-Value Analysis and Imputation

Missing-value analysis is performed for the selected parameters before WQI construction.

Missing individual measurements do not automatically exclude an observation row or a station-year group. In the WQI builders, availability is assessed within each StationCode-Year group: a parameter is available if it has at least one non-missing measurement. Missing measurements are omitted separately for each parameter, rather than applying complete-case deletion across the five WQI parameters. The baseline all-six-missing row filter is applied before this grouped assessment, as described above.

3.3.1 Missing-Value Imputation

In the missingness analysis, two complementary imputation strategies were investigated to reduce the loss of usable parameter information. The first strategy was parameter-based predictive imputation, in which available water-quality parameters and temporal information were used to estimate missing measurements. The second strategy was station-based temporal imputation, which used the temporal behavior of measurements within

the same monitoring station to estimate missing values.

The final imputation implementation uses Random Forest regression for BOD, Ammonia-N, and Nitrite-N, followed by station-wise adjacent-observation averaging for Nitrate-N and pH. Neither median replacement nor KNN imputation is used in this final preprocessing script. DO is not imputed by this script.

For each regression target, training rows must contain an observed target and nonmissing values for all its predictors. The regressor is instantiated with `n_estimators=100`, `random_state=42`, and `n_jobs=-1`; no other hyperparameters are explicitly set. Predictions replace only missing target values for rows with complete predictor values. The targets are processed sequentially in the order BOD, Ammonia-N, and Nitrite-N, so a later imputation stage can use values recovered by an earlier stage. Observed values are not overwritten.

For Nitrate-N and pH, observations are sorted by StationCode and Date. Within each station, the immediately preceding and immediately following values are obtained using `shift(1)` and `shift(-1)`. A missing value is replaced with the arithmetic mean of these two values only when both are available. This is an adjacent-record average, not time-distance-weighted interpolation. Missing values without two available adjacent observations remain missing; no extrapolation or iterative filling is implemented. Each parameter has a binary imputation flag identifying replaced measurements.

The two approaches were evaluated independently using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2. The validation results showed that the performance of the approaches varied across parameters. Therefore, a parameter-specific strategy was considered rather than applying a single imputation method uniformly to all variables.

3.4 CCME-WQI Construction

3.4.1 Selection of WQI Formulation

A Water Quality Index provides a numerical representation of the overall water-quality condition by combining measurements from multiple parameters according to a defined methodology.

Because different WQI formulations use different parameters, reference objectives, normalization methods, and aggregation procedures, the selection of the WQI formulation is a very important part of the study.

The Canadian Council of Ministers of the Environment Water Quality Index (CCME-WQI) framework combines multiple parameters and repeated observations through scope (F1), frequency (F2), and amplitude (F3) [12]. This method reference establishes the formulation, not the statutory provenance of the study-specific objectives.

3.4.2 Parameter Selection

For the final WQI construction, five parameters are selected:

These parameters represent different aspects of the current river-water condition, including organic pollution, nutrient pollution, and acid-base characteristics, among others.

DO is included among the six ML predictors but excluded from the five-parameter WQI target. The uploaded construction scripts define objectives only for BOD, Ammonia-N, Nitrite-N, Nitrate-N, and pH; they do not establish an additional regulatory or measurement-representation rationale for excluding DO.

3.4.3 Selected Parameters

Five parameters were used to construct the WQI:

- BOD
- Ammonia-N
- Nitrite-N
- Nitrate-N
- pH

These parameters were selected based on availability in the NIEA dataset and their relevance to the assessment of organic pollution, nutrient contamination, and general chemical condition.

Table 3 lists the study-defined water-quality objectives used for CCME-WQI construction and guideline-exceedance analysis, together with the minimum-data criteria documented in the uploaded WQI construction scripts [13]. These are implemented study objectives, not verified statutory limits or a claim that CCME prescribes these particular values. Their original scientific or regulatory provenance was not established in the reviewed project materials; therefore, they are not presented as statutory or legally binding limits.

Table 1 Selected parameter-specific imputation approaches

Parameter	Selected approach
BOD	Random Forest regression using Ammonia-N, Nitrite-N, Nitrate-N, pH, Year, and Month.
Ammonia-N	Random Forest regression using BOD, Nitrite-N, Nitrate-N, pH, Year, and Month.
Nitrite-N	Random Forest regression using BOD, Ammonia-N, Nitrate-N, pH, Year, and Month.
Nitrate-N	Arithmetic mean of the immediately preceding and following nonmissing adjacent observations in the same station.
pH	Arithmetic mean of the immediately preceding and following nonmissing adjacent observations in the same station.

Table 2 Parameters selected for WQI construction

Parameter	Description
BOD	Biochemical Oxygen Demand
Ammonia-N	Ammonia Nitrogen
Nitrite-N	Nitrite Nitrogen
Nitrate-N	Nitrate Nitrogen
pH	Acidity or alkalinity indicator

Table 3 WQI thresholds and minimum-data criteria

Parameter / Criterion	Threshold / Requirement
BOD	3.0 mg/L
Ammonia-N	0.05 mg/L
Nitrite-N	0.5 mg/L
Nitrate-N	0.5 mg/L
pH	6.2–7.3
Minimum Station-Year observations	10
Minimum available WQI parameters	4/5

3.4.4 Station-Year Aggregation

Both WQI builders group individual observations by StationCode and Year. The 175,615 baseline core rows form 15,502 groups, whereas the 178,680 imputed-input rows form 16,639 groups. The supplied inputs contain no missing StationCode or Year values, so no rows are lost through missing grouping keys. These group counts precede eligibility filtering and must not be confused with the number of final WQI records. The baseline group-size median is 12 and the mean is approximately 11.33.

The minimum observation requirement is implemented as `len(group) >= 10`. It counts rows in the input group, not complete measurement vectors or valid tests for an individual parameter. Consequently, a retained row with all WQI measurements missing still contributes to the imputed-input observation count, although it contributes no parameter tests. A group must also contain at least four available parameters out of BOD, Ammonia-N, Nitrite-N, Nitrate-N, and pH. Availability requires at least one nonmissing value for that parameter within the group. DO does not contribute to WQI parameter availability. The builders additionally reject a group if the total number of available tests is zero; no separate zero-test exclusion remains after the four-parameter requirement in these inputs.

The five WQI parameters are not reduced to medians or means before computing the index. For each eligible group, the builder iterates over all nonmissing measurements of each parameter. Each available measurement contributes one test, and each guideline failure contributes a failed test and an excursion. A parameter is counted as failed if any of its available measurements fails its objective. F1 is computed from failed and available parameter counts, F2 from failed and total test counts, and F3 from the sum of excursions divided by total tests. Across the eligible groups, the baseline branch contains 741,697 available parameter tests and the imputed branch contains 750,603. Observation-row counts and parameter-test counts are therefore different quantities.

The baseline builder combines F1, F2, and F3 using the normalized CCME-WQI expression and writes one row per eligible station-year, with StationCode, Year, Observations, Parameters, F1, F2, F3, CCME_WQI, and Category. Category is assigned by descending cutoffs of 95, 80, 65, and 45. The saved imputed output has the same columns.

The separate exploratory station-year script writes 15,502 groups using parameter medians and the first Location, PrimaryBasin, and GridReference values. That summary file is not the input to either WQI builder. Likewise, the ML-preparation script computes six station-year predictor means separately; those means are not used to calculate the WQI target.

3.4.5 CCME-WQI Components

The CCME-WQI uses three main factors.

F1 — Scope

F1 represents the proportion of selected parameters that fail the water-quality objectives at least once during the assessment period.

It can be expressed as:

$$F_1 = \frac{N_{\text{failed parameters}}}{N_{\text{available parameters}}} \times 100 \tag{1}$$

A higher F1 indicates that a larger proportion of the assessed parameters have exceeded their respective objectives.

F2 — Frequency

F2 represents how frequently individual measurements fail the objectives.

$$F_2 = \frac{N_{\text{failed tests}}}{N_{\text{total tests}}} \times 100 \tag{2}$$

This component considers the repeated nature of the monitoring observations. Therefore, a parameter that exceeds its objective repeatedly will contribute more strongly to the frequency component.

F3 — Amplitude

F3 represents the magnitude by which failed observations exceed their corresponding objectives.

For each failed observation, an excursion is calculated. For an upper-limit objective:

$$\text{Excursion} = \frac{\text{Value}}{\text{Objective}} - 1 \tag{3}$$

Excursion — lower pH limit

$$\text{Excursion} = \frac{\text{Lower Limit}}{\text{Value}} - 1 \tag{4}$$

Excursion — upper pH limit

$$\text{Excursion} = \frac{\text{Value}}{\text{Upper Limit}} - 1 \tag{5}$$

The normalized sum of excursions is then used to calculate F3.

$$\text{nse} = \frac{\sum \text{Excursions}}{N_{\text{total tests}}} \tag{6}$$

$$F_3 = \frac{\text{nse}}{0.01\text{nse} + 0.01} \tag{7}$$

In the implemented calculation, the normalized excursion component is used to ensure that the final WQI remains within the standard 0–100 scale.

Table 4 CCME-WQI classification

CCME-WQI	Category
95–100	Excellent
80–94	Good
65–79	Fair
45–64	Marginal
0–44	Poor

The same CCME-WQI formulation and water-quality objectives were applied consistently to both the baseline and imputed datasets to ensure that differences in subsequent modelling results could be attributed to the preprocessing strategy rather than changes in the WQI formulation.

3.4.6 CCME-WQI Calculation

The three factors are combined using the CCME-WQI formulation:

$$\text{CCME-WQI} = 100 - \frac{\sqrt{F_1^2 + F_2^2 + F_3^2}}{1.732} \tag{8}$$

The resulting value ranges from 0 to 100, with higher values indicating better water-quality conditions.

3.4.7 WQI Classification

3.5 Predictor and Target Definition

3.5.1 WQI as the ML Target

The calculated CCME_WQI is used as a continuous regression target for the subsequent predictive models.

The input features consist of selected water-quality measurements, while the WQI category is not used as an input feature because the category is derived directly from the WQI value.

The modelling relationship can be represented as:

BOD
DO
Ammonia-N
Nitrite-N
Nitrate-N
pH
↓
Machine Learning / Deep Learning
↓
Predicted CCME-WQI

This helps prevent direct target leakage and allows the models to learn the relationship between the underlying water-quality parameters and the constructed WQI.

The same feature-target relationship and modelling procedure are applied independently to the baseline and imputed WQI datasets. This enables a controlled comparison of model performance before and after missing-value recovery.

3.5.2 Problem Formulation

The Water Quality Index prediction task was formulated as a supervised machine-learning regression problem. The objective was to predict the continuous CCME Water Quality Index (CCME-WQI) value from the selected water-quality parameters.

The input features used for the model development were:

- Biochemical Oxygen Demand (BOD)
- Dissolved Oxygen (DO)
- Ammonia Nitrogen (Ammonia-N)
- Nitrite Nitrogen (Nitrite-N)
- Nitrate Nitrogen (Nitrate-N)
- pH

It should be noted that Dissolved Oxygen was not used in the calculation of the CCME-WQI target. The CCME-WQI in this study was constructed using the five selected parameters: BOD, Ammonia-N, Nitrite-N, Nitrate-N, and pH. However, DO was included as an additional predictor during machine-learning model development because it is an important water-quality parameter and may provide supplementary information for predicting the overall water-quality condition. Thus, DO was used as an input feature for WQI prediction but was not used to calculate the WQI target.

The target variable was the calculated CCME-WQI.

Two datasets were used for model development and comparison:

- The baseline dataset has 12,497 station-year records.
- The imputed dataset has 12,503 station-year records.

The objective was to compare the predictive performance of multiple machine-learning algorithms on both datasets.

3.5.3 Machine Learning Dataset Preparation

Before model training, both datasets were examined for missing values and duplicate records. No missing values were found in the final machine-learning datasets. Duplicate StationCode-Year combinations were also checked, and no duplicate records were identified.

The final baseline dataset contained 12,497 records, while the imputed dataset contained 12,503 records. Each record represented a station-year observation and contained the six predictor variables, the CCME-WQI target value, and the corresponding water-quality category.

The machine-learning prediction task used the six selected water-quality parameters as input variables. StationCode, Year, and Category were not used as predictive input features.

The model-preparation script computes the arithmetic mean of each of the six predictors within each StationCode-Year group in the core dataset. These same core-derived predictor means are inner-joined separately with the baseline and imputed WQI targets on StationCode and Year. The supplied implementation does not recompute ML predictor means from the imputed raw measurements; the two modelling datasets differ through their WQI targets and eligible station-year joins. Missing values are excluded from each parameter mean by the aggregation operation. The DNN and TabNet scripts additionally remove any rows with missing predictor or target values before splitting.

The six predictor means are constructed from the 175,615-row core dataset in both branches. Each of the 12,497 baseline WQI keys and 12,503 imputed WQI keys has a corresponding core-derived predictor group, so the inner joins do not further reduce either saved WQI record count. The resulting saved ML datasets contain 12,497 and 12,503 rows, respectively, with no missing values in the six predictors or target and no duplicate StationCode-Year keys. Consequently, the DNN/TabNet missing-value checks do not reduce these supplied datasets. The six additional imputed-target keys have core predictor groups even though those groups contain only nine core observations and were excluded from baseline WQI construction.

The target is the constructed station-year CCME-WQI based on BOD, Ammonia-N, Nitrite-N, Nitrate-N, and pH. The six predictors are BOD, DO, Ammonia-N, Nitrite-N, Nitrate-N, and pH, with DO serving only as an additional predictor. The implementation estimates the index for the represented station-year from its predictor means; it does not define a future target, lagged forecasting task, or forecast horizon.

3.5.4 Input Features and Target

The six input features used by all models are:

1. BOD
2. DO
3. Ammonia-N
4. Nitrite-N
5. Nitrate-N
6. pH

The station-year CCME-WQI was used as the prediction target. All models were evaluated on both the baseline and imputed datasets to determine whether the missing-value treatment had an observable effect on prediction performance.

3.6 Machine Learning Models

Random Forest [4] and Extra Trees combine predictions from multiple decision trees, with Extra Trees introducing greater randomness during tree construction.

Gradient Boosting builds trees sequentially, with each new tree focusing on reducing the errors made by previous trees. XGBoost provides an optimized implementation of gradient boosting for structured data [3]. CatBoost is another boosting-based method included in the comparative evaluation [2].

3.6.1 Implemented Machine-Learning Configurations

Each ML model is fitted separately on the baseline and imputed training subsets using the six predictors. The scripts pass the predictor values directly to the estimators, without a scaling stage. Table 5 lists every explicitly supplied constructor argument. Parameters not specified in these scripts are left to the installed library defaults; their numerical values and library versions are not recorded in the supplied implementation and are not inferred here.

3.7 Deep Learning Models

TabNet was included as an attention-based neural model for tabular data [5].

3.7.1 Implemented DNN Architecture and Training

The DNN uses a TensorFlow/Keras Sequential architecture with a six-feature input, four Dense hidden layers containing 128, 64, 32, and 16 units, respectively, each with ReLU activation, and a single Dense output unit without an explicitly specified activation. No dropout, batch-normalization layer, or explicit regularization is included. A StandardScaler is fitted only on the training predictors and is then applied to the test predictors; the target is not scaled.

The model is compiled with `optimizer="adam"`, `loss="mse"`, and `metrics=["mae"]`. Training uses `validation_split=0.20`, `epochs=100`, `batch_size=32`, and `verbose=1`. The validation fraction is taken from the training subset rather than from the holdout test subset. No callbacks, early stopping, explicit optimizer learning rate, TensorFlow/NumPy seed, or explicit training-shuffle argument are specified. Consequently, the fixed split seed does not establish seeded neural-network initialization. Predictions are generated with `verbose=0` and flattened before metric calculation.

3.7.2 Implemented TabNet Architecture and Training

The TabNetRegressor is instantiated with `n_d=16`, `n_a=16`, `n_steps=5`, `gamma=1.5`, `lambda_sparse=1e-4`, `optimizer_fn=torch.optim.Adam`, `seed=42`, and `verbose=10`. Other architecture and optimizer settings, including the optimizer learning rate, are not explicitly configured. A StandardScaler is fitted on the training predictors and applied to the test predictors, after which both predictor arrays are converted to `float32`. Targets are reshaped into single-column arrays and are not scaled.

Training specifies `max_epochs=100`, `patience=15`, `batch_size=256`, `virtual_batch_size=128`, `num_workers=0`, and `drop_last=False`. The evaluation sets are the training subset and the same 20% test subset, named `train` and `test`, with `eval_metric=["mae"]`. The test subset is thus

Table 5 Explicitly implemented machine-learning configurations

Model	Explicit configuration
Random Forest	`n_estimators=100`, `random_state=42`, `n_jobs=-1`.
Extra Trees	`n_estimators=100`, `random_state=42`, `n_jobs=-1`.
Gradient Boosting	`n_estimators=100`, `learning_rate=0.1`, `max_depth=3`, `random_state=42`.
XGBoost	`n_estimators=300`, `learning_rate=0.05`, `max_depth=6`, `subsample=0.8`, `colsample_bytree=0.8`, `objective="reg:squarederror"`, `random_state=42`, `n_jobs=-1`.
CatBoost (holdout)	`iterations=300`, `learning_rate=0.05`, `depth=6`, `loss_function="RMSE"`, `verbose=0`, `random_seed=42`.

used for early-stopping evaluation as well as final performance reporting; it is not an independent untouched test set for this implementation. No separate validation split or explicit training-shuffle argument is supplied.

3.8 Experimental Setup and Data Splitting

Both datasets were divided into training and testing subsets using an 80:20 ratio. A fixed random state of 42 was used to ensure the reproducibility of the experiments.

All five ML scripts and both DL scripts call `train_test_split` with `test_size=0.20` and `random_state=42`. No station grouping, chronological separation, or stratification is supplied. The split is therefore a random station-year row split, not a held-out-station or future-year evaluation.

For the baseline dataset:

- Training samples: 9,997
- Testing samples: 2,500

For the imputed dataset:

- Training samples: 10,002
- Testing samples: 2,501

The same data-splitting strategy was applied to all machine-learning models to ensure a fair comparison.

3.9 Model Evaluation

Mean Absolute Error (MAE): Measures the average absolute difference between the predicted and actual WQI values.

Root Mean Squared Error (RMSE): Measures the prediction error while giving greater weight to larger errors.

Coefficient of Determination (R^2): Indicates how well the model explains the variation in the target variable.

3.9.1 Model Comparison and Selection

The ML and DL models were compared using both the baseline and imputed datasets. Based on the evaluation metrics, the CatBoost model was selected for further stages of the study because of its strong predictive performance. Five-fold cross-validation was then applied to the selected CatBoost model to assess the consistency of its performance across different subsets of the data. Separate CatBoost fits are used in the holdout comparison, five-fold validation, SHAP analysis, and recommendation modules; the validation fit is not assumed to be the same fitted artifact used in all subsequent components.

3.10 CatBoost Cross-Validation

Cross-validation is performed independently on each complete ML dataset, not only on the 80% holdout-training subset. The splitter is `KFold(n_splits=5, shuffle=True, random_state=42)`. For each fold, positional indices select four folds for training and the remaining fold for evaluation, and a new CatBoostRegressor is fitted. Its configuration is `iterations=300`, `depth=6`, `learning_rate=0.05`, `loss_function="RMSE"`, `random_seed=42`, and `verbose=False`.

Each fold yields MAE; RMSE, calculated as the square root of mean squared error; and R^2. The script summarizes the five-fold values with `np.mean` and the sample standard deviation using `np.std(ddof=1)`. It does not group folds by station or order them chronologically. This procedure measures variation across shuffled station-year row partitions; it does not establish generalization to previously unseen stations, external basins, or future years.

3.11 Explainable AI Using SHAP

SHapley Additive exPlanations (SHAP) describe feature contributions to the fitted CatBoost predictions of the constructed station-year CCME-WQI [1]. The six explained predictors are BOD, DO, Ammonia-N, Nitrite-N, Nitrate-N, and pH. Global explanations summarize contributions across evaluated samples, whereas local explanations describe the additive contributions to one individual prediction. Neither type identifies a physical cause of water-quality change.

The figures in this section are saved outputs of `04_SHAP/01_shap_analysis.py` and `04_SHAP/02_local_shap_analysis.py`. Both scripts exclude records missing any predictor or target, use an 80:20 train–test split with `random_state=42`, fit CatBoost with 300 iterations, depth 6, learning rate 0.05, RMSE loss, and `random_seed=42`, and compute `shap.TreeExplainer` attributions on the test predictors.

[Station-Year Predictors]
↓
[CatBoost Regressor] ⟶ Estimates Constructed CCME-WQI
↓
[TreeSHAP Explainer] ⟶ Global Contributions & Local Attributions
↓
[Decision Logic] ⟶ Cross-references Guideline Thresholds
[Decision-Support Outputs] ⟶ Priority, Advice & Possible Pressures

3.11.1 Local Sample-Level Decomposition

For an individual station-year predictor vector x, local TreeSHAP decomposes the fitted prediction into the explainer expected output and six feature contributions:

$$\hat{y}(x) = \phi_0 + \sum_{i=1}^{M} \phi_i(x).$$

Here $M = 6$, ϕ_0 is returned by `explainer.expected_value`, and $\phi_i(x)$ is the signed attribution to predictor i. The expected model output is not assumed to equal the observed target mean. A positive attribution pushes the prediction upward relative to this reference; a negative attribution pushes it downward. These signs do not by themselves establish improvement or deterioration of the actual water body.

3.11.2 Bridging Interpretability to Field Decision Support

The separate recommendation implementation uses local negative attributions to select parameter-specific advisory text, as documented in Section 3.12. Guideline exceedances remain measurement-threshold checks, while priority is determined by predicted category and exceedance count. The plots presented here use test-set explanation fits; they should not be conflated with the full-data fit used to generate the recommendation CSV outputs.

Interpretation boundary.

SHAP quantifies feature attribution within a fitted regressor. A negative BOD attribution means that its allocated contribution lowers the predicted index relative to the explainer reference; it does not verify organic-pollution causation or identify a discharge source. Possible pressure messages are predefined interpretations, not causal findings.

3.12 Decision-Support Recommendation Framework

3.12.1 Model Prediction and Water-Quality Category

The module processes the baseline and imputed machine-learning tables independently. It requires the six predictors BOD, DO, Ammonia-N, Nitrite-N, Nitrate-N, and pH, together with the constructed `CCME_WQI` target. Records with a missing value in any of these seven columns are excluded from this module. A separate CatBoost regressor is fitted to all remaining records in each table, saved as a dataset-specific model, and used to predict those same records. Thus, the recommendation outputs generated by this script are in-sample decision-support outputs, not a new holdout evaluation or a forecast of future water quality.

The predicted category, stored as `Predicted_Category`, is assigned by testing thresholds in descending order: Excellent for a prediction of at least 95; Good for a prediction of at least 80 but below 95; Fair for a prediction of at least 65 but below 80; Marginal for a prediction

of at least 45 but below 65; and Poor for a prediction below 45. The category function returns Unknown for a missing prediction. The script does not clip predictions before categorization.

3.12.2 Guideline Checks and Parameters Requiring Attention

Table 6 records the numerical conditions implemented in the guideline dictionary. A measurement equal to a boundary is within the implemented guideline. Missing values receive the status Missing, although the complete-case selection above removes records with missing predictors before recommendations are generated.

For DO, this status is a software fallback because no DO guideline exists in the dictionary, not evidence of compliance with an oxygen standard.

For each record, the script constructs an internal `exceeded_parameters` list by checking the guideline-defined parameters in the order BOD, Ammonia-N, Nitrite-N, Nitrate-N, and pH. A parameter is appended once if its status is Above guideline or Below guideline. The list therefore identifies the parameters requiring attention on the basis of threshold violations, and its length counts distinct exceeded parameters rather than individual measurement tests. DO is not included in this list.

3.12.3 Exact Monitoring-Priority Rules

Let E denote the number of parameters in the internal exceedance list. The priority function applies the following rules in order, returning as soon as a condition is satisfied:

1. A predicted category of Poor produces HIGH priority.
2. A predicted category of Marginal produces HIGH priority.
3. Any remaining category with $E \geq 2$ produces HIGH priority.
4. A remaining predicted category of Fair produces MEDIUM priority.
5. All remaining cases produce LOW priority.

The generic fallback also assigns LOW to Unknown when $E < 2$. SHAP magnitudes, the selected advice parameter, and the actual category do not enter this priority function. In particular, a single exceedance does not automatically imply MEDIUM or HIGH priority.

3.12.4 SHAP-Based Advice Selection and Pressure Interpretation

The script computes local attributions for all six predictors using `shap.TreeExplainer` on the fitted CatBoost model. For each record, it collects features with strictly negative SHAP values and sorts them by their signed values in ascending order. If this collection is nonempty, the feature with the most negative SHAP value becomes the selected advice parameter, and its attribution is recorded. Equal negative values retain the predictor order BOD, DO, Ammonia-N, Nitrite-N, Nitrate-N, and pH.

If no negative attribution exists but the exceedance list is nonempty, the first parameter in that list is selected and its recorded SHAP value is missing. If neither condition applies, the selected parameter is None and its recorded SHAP value is missing.

Table 7 reproduces the implemented mappings. This is deterministic selection of predefined text, not free-form text generation. A selected negative SHAP contributor need not exceed its guideline and need not be a member of the threshold-based attention list. DO can therefore select oxygen-related advice through SHAP, despite having no implemented guideline threshold or role in the exceedance count.

When the selected parameter has no dictionary entry, the pressure output is "No major pollution pressure identified by the rule-based system" and the recommendation is "Continue routine monitoring and maintain current water-quality surveillance."

The pressure messages are predefined hypotheses for follow-up, not findings of source attribution.

3.13 Streamlit Dashboard Implementation

It accepts six predictor values and estimates the constructed station-year CCME-WQI; it does not implement temporal forecasting or demonstrate a particular inference latency. Inputs should be interpreted in the context of the station-year mean predictors used for model development. The

Table 6 Implemented parameter-level guideline conditions in the recommendation module.

Parameter	Exceedance condition	Status assigned
BOD	Value $>$ 3.0 mg/L	Above guideline
Ammonia-N	Value $>$ 0.05 mg/L	Above guideline
Nitrite-N	Value $>$ 0.5 mg/L	Above guideline
Nitrate-N	Value $>$ 0.5 mg/L	Above guideline
pH	Value $<$ 6.2	Below guideline
pH	Value $>$ 7.3	Above guideline
DO	No threshold implemented	Within guideline for a nonmissing value

Table 7 Exact parameter-specific pressure and recommendation messages implemented in the module.

Selected parameter	Possible pressure text	Recommendation text
BOD	Possible organic pollution or wastewater-related pressure	Investigate possible organic or wastewater pollution sources and increase monitoring.
Ammonia-N	Possible sewage, organic-waste, or nutrient pollution pressure	Investigate possible sewage or organic-waste inputs and increase monitoring of nitrogen pollution.
Nitrite-N	Possible nitrogen-cycle disturbance or wastewater-related pressure	Investigate possible wastewater or nitrogen-cycle disturbances and monitor nitrogen compounds.
Nitrate-N	Possible nutrient runoff, agricultural, or wastewater-related pressure	Investigate possible nutrient runoff or wastewater sources and strengthen nutrient monitoring.
pH	Possible chemical or industrial influence	Investigate possible chemical or industrial influences and verify pH measurements.
DO	Possible oxygen-depletion or organic-pollution stress	Increase monitoring for oxygen depletion and investigate possible organic-pollution sources.

dashboard displays a clipped prediction and category, configured guideline checks, local SHAP contributions, and advisory outputs.

3.13.1 Core System Modules

1. Interactive Field Input Panel: Environmental officers input the 6 parameter measurements through sliders or direct numerical fields with built-in range validation.
2. WQI Estimation: The dashboard loads the dataset-specific baseline or imputed CatBoost artifact and estimates the constructed station-year CCME-WQI from six input values. No inference-time benchmark is supplied.
3. Configured Guideline Checks: The dashboard compares the five guideline-defined parameters with the study-defined water-quality objectives and displays their statuses in a table. DO has no configured threshold. These checks do not establish statutory compliance.
4. Local Model Explanation: The dashboard displays signed SHAP contributions in a bar chart and table relative to the explainer expected output. The uploaded dashboard script does not generate waterfall or force plots; the manuscript waterfall figures come from the separate local-analysis script.
5. Advisory Outputs: The dashboard generates advisory outputs based on two primary conditions: the predicted category and whether study-defined objectives have been exceeded. In parallel, the offline Recommendation Engine applies local SHAP attributions to identify a parameter-specific advisory message whenever a negative attribution is present. Although both components are guided by the same overarching monitoring-priority framework, the procedures used to determine the final advisory output are implemented independently.

4 Results

4.1 Dataset and Missing-Value Results

The processed dataset has the following missing observations:

Following imputation, a validation step was performed to confirm that the process had not produced any negative values. The resulting parameter ranges and summary statistics were subsequently examined in relation to the available measurements to evaluate the characteristics of the completed dataset prior to WQI construction. Importantly, this validation was intended to

Table 8 STREAMLIT APPLICATION WORKFLOW

1. Field Input Layer	Sliders / number fields for BOD, DO, NH3-N, NO2-N, NO3-N, and pH.
2. Inference Engine	Passes vector to serialized CatBoost model artifact.
3. Target Output	Displays CCME-WQI score [0–100] & quality tier.
4. Guideline Checker	Compares values against study-defined water-quality objectives.
5. SHAP Visualizer	Displays signed local SHAP contributions as a bar chart and table.
6. Triage Advisory	Displays dashboard monitoring priority and follow-up advice.

Table 9 Missing observations in the processed dataset

Parameter	Missing observations	Missing (%)
BOD	24,351	13.87%
DO	24,578	14.00%
Ammonia-N	948	0.54%
Nitrite-N	11,153	6.35%
Nitrate-N	11,743	6.69%
pH	20,314	11.57%

assess the behavior and plausibility of the imputed observations rather than to demonstrate formal equivalence between the imputed and observed distributions.

4.2 WQI Results

4.2.1 Final WQI Dataset

The saved baseline and imputed WQI datasets contain 12,497 and 12,503 station-year rows, respectively. Both originate from the same raw source file, but the baseline branch first excludes 3,065 all-six-missing rows and the imputed branch does not. Table 10 distinguishes raw rows, builder-input rows, eligible-group observation rows, and final station-year records. Reapplying the eligibility conditions to the supplied inputs reproduces the saved output keys and the summed Observations columns exactly.

The saved imputed WQI keys contain all 12,497 baseline keys and six additional keys. Each additional group contains nine baseline core rows but ten rows in the original and imputed inputs, and all five WQI parameters are already available before imputation. The six keys are UKGBNIF10325–2003, UKGBNIF10326–2003, UKGBNIF10361–2020, UKGBNIF10363–2020, UKGBNIF10907–2007, and UKGBNIF11451–2020. The six-record increase is therefore explained by retention of an additional all-six-missing row in each group, allowing it to meet the ten-row criterion; it is not evidence that imputation made an additional WQI parameter available in these groups. Applying the same eligibility rules to the original, unimputed raw observations also yields 12,503 eligible groups.

The two saved WQI datasets are retained for the existing modelling comparison. Their preparation differs in both row-retention rules and measurement recovery, so the six-record difference should not be attributed solely to successful imputation.

The CCME-WQI calculation was performed independently for both the baseline and imputed datasets using the same formulation and classification thresholds. The baseline dataset produced 12,497 station-year WQI records, while the imputed dataset produced 12,503 station-year WQI records. No calculated WQI value was below 0 or above 100 during the validation process.

For the baseline dataset, the WQI values ranged from 7.07 to 100, with a mean of 49.43 and a median of 46.66. For the imputed dataset, the WQI values ranged from 7.52 to 100, with a mean of 49.36 and a median of 46.58.

The resulting distribution in the current dataset is:

The WQI category distributions of the baseline and imputed datasets remained broadly similar. The similarity between the two distributions indicates that the imputation process did not substantially alter the overall WQI classification pattern.

4.2.2 Exploratory Analysis for Machine Learning

For the baseline dataset, the mean CCME-WQI was 49.43, with a standard deviation of 15.02. For the imputed dataset, the mean CCME-WQI was 49.36, with a standard deviation of 15.02.

The imputed dataset showed nearly identical relationships between the predictor variables and the target variable.

4.3 Machine Learning and Deep Learning Results

The benchmark table reports prediction errors and coefficients of determination, but no statistical significance test is documented. Small branch-level metric differences do not by themselves

Table 10 Observation and station-year counts through dataset construction

Stage / Count unit	Baseline	Imputed
Original raw observation rows	178,680	178,680
Rows excluded by all-six-missing preprocessing filter	3,065	0
Observation rows supplied to WQI builder	175,615	178,680
Station-year groups before eligibility checks	15,502	16,639
Groups excluded for fewer than ten input rows	2,327	3,452
Groups excluded for insufficient WQI parameters after passing the row-count check	678	684
Observation rows within eligible groups	157,701	158,965
Final station-year WQI records	12,497	12,503

Table 11 WQI category distributions

Category	Baseline	Imputed
Poor	5,691	5,723
Marginal	5,044	5,021
Fair	1,189	1,190
Good	554	550
Excellent	**19**	19
Total	**12,497**	12,503

Table 12 Baseline feature correlations with CCME-WQI

Feature	Correlation with CCME-WQI
Nitrate-N	-0.6997
Nitrite-N	-0.6682
Ammonia-N	-0.3407
pH	-0.2953
BOD	-0.2930
DO	0.2895

establish preservation of every feature distribution.

The uploaded project RMSE comparison visualization is reproduced in Figure 2.

Figures 3 and 4 reproduce the baseline and imputed actual-versus-predicted plots generated by `10_prediction_analysis.py`. They visualize prediction agreement on the specified random splits, not geographic transfer or future forecasting.

4.4 CatBoost Cross-Validation Results

Baseline Set: Mean MAE = 2.9960 ± 0.0252, Mean RMSE = 4.0448 ± 0.0356, Mean $R^2 = 0.9274 \pm 0.0018$.

Imputed Set: Mean MAE = 3.0042 ± 0.0235, Mean RMSE = 4.0463 ± 0.0376, Mean $R^2 = 0.9273 \pm 0.0031$.

4.5 SHAP Explainability Results

4.5.1 Global Contributions Across Evaluated Samples

Global importance is calculated as the mean absolute SHAP value for each predictor over the evaluated test samples: 2,500 baseline records and 2,501 imputed records, as verified in the saved SHAP-value tables. Figure 5 presents these aggregate magnitudes, while Figure 6 shows the distribution of signed contributions and corresponding predictor values. Mean absolute importance measures attribution magnitude, not its direction or environmental importance.

The summary plots show higher nitrate and ammonia values to be predominantly associated with negative contributions, and lower values to be predominantly associated with positive contributions in these evaluated samples. DO has the lowest mean absolute attribution in both tables; this does not establish that oxygen is environmentally unimportant.

4.5.2 Global SHAP Feature Hierarchy

The saved mean absolute SHAP tables for the baseline and imputed test samples support the following ranking of model attribution, shown in Figure 5:

$$\text{Nitrate-N} > \text{Ammonia-N} > \text{BOD} > \text{Nitrite-N} > \text{pH} > \text{DO}$$

Nitrate-N and Ammonia-N have the highest mean absolute SHAP values, followed by BOD, in both saved outputs. This ordering describes contributions to the fitted predictions over the evaluated samples, not causal drivers of environmental conditions. DO has the lowest mean absolute attribution, which does not imply environmental

Table 13 Comparative Benchmark Evaluation

Model Family	Algorithm	Baseline MAE	Baseline RMSE	Baseline R^2	Imputed MAE	Imputed RMSE	Imputed R^2
Tree Ensembles	Random Forest	3.0245	4.1346	0.9231	3.0870	4.2095	0.9204
	Extra Trees	3.0872	4.1626	0.9221	3.1329	4.1942	0.9210
Boosting	Gradient Boosting (GBDT)	3.0791	4.1093	0.9240	3.1068	4.1384	0.9231
	XGBoost	2.9187	4.0063	0.9278	2.9579	4.0368	0.9268
	CatBoost (Selected)	2.9632	3.9849	0.9286	2.9799	4.0024	0.9280
Deep Learning	Deep Neural Net (DNN)	3.3088	4.4594	0.9105	3.4729	4.5312	0.9078
	TabNet	4.8551	6.3383	0.8193	4.6399	6.0572	0.8352

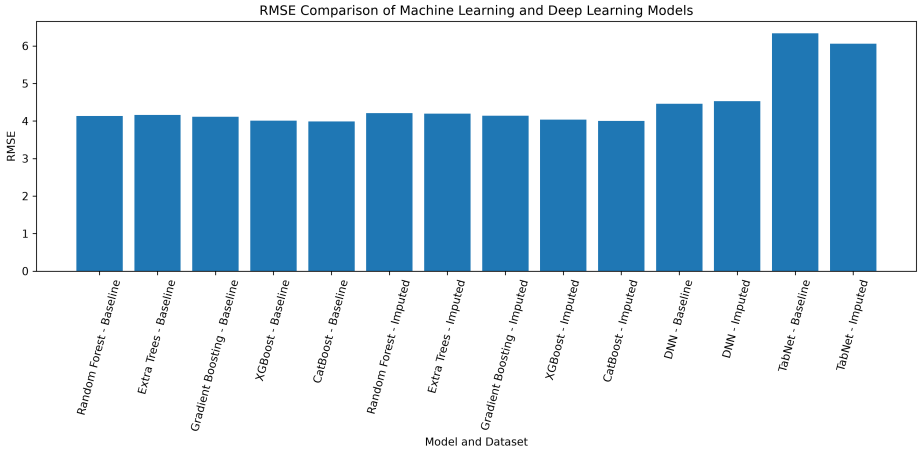


Fig. 2 Original project RMSE comparison across the five machine-learning models, DNN, and TabNet for baseline and imputed datasets.

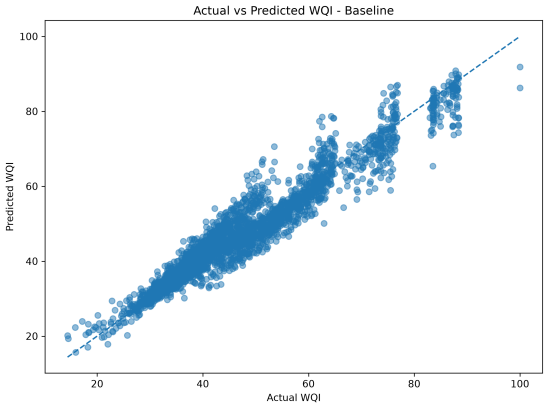


Fig. 3 Constructed station-year CCME-WQI versus CatBoost prediction for the baseline test set. The dashed line is the identity line; deviations indicate prediction error for individual held-out records.

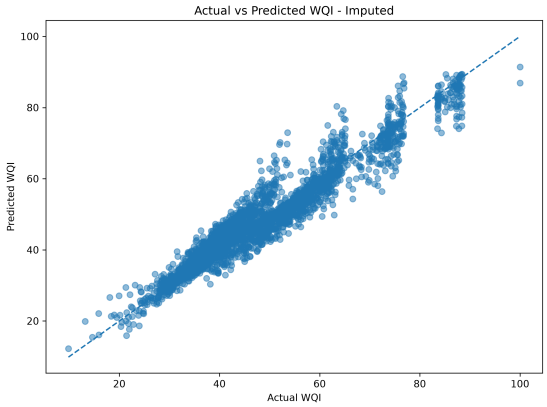


Fig. 4 Constructed station-year CCME-WQI versus CatBoost prediction for the imputed test set. The dashed line is the identity line. The plotted target is the constructed station-year index, not an independently measured future water-quality outcome.

unimportance. The signed distributions and individual explanations are provided in Figures 6, 8, and 9.

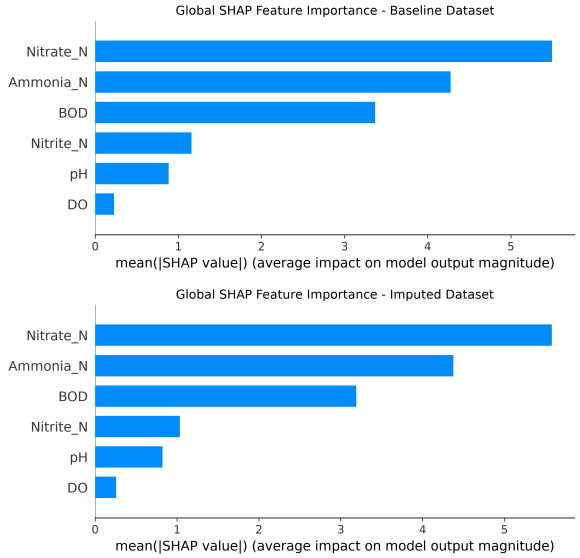


Fig. 5 Global mean absolute SHAP importance for the baseline (top) and imputed (bottom) CatBoost explanation models, evaluated over 2,500 and 2,501 test records, respectively. Bar lengths summarize contribution magnitudes in predicted WQI units and do not indicate causal effects.

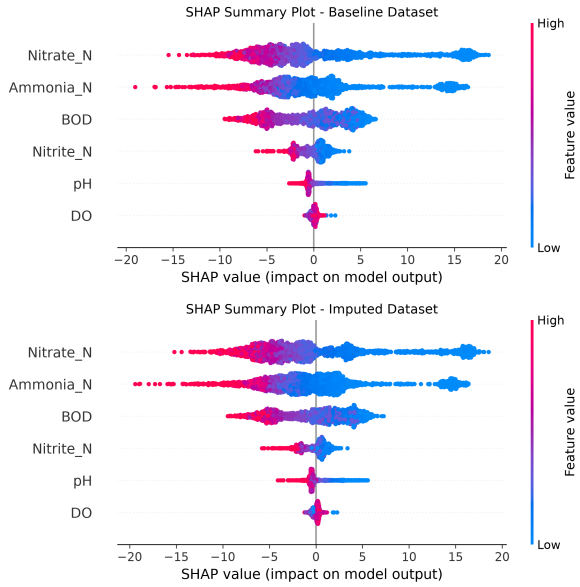


Fig. 6 Global SHAP summary plots for baseline (top) and imputed (bottom) test records. Each point represents one sample-level attribution; horizontal position indicates the signed contribution relative to the model expected output, and color indicates the relative predictor value. Positive contributions push the prediction upward and negative contributions push it downward.

4.5.3 Nitrate Dependence in the Fitted Models

Figure 7 provides a complementary view of the highest-ranked predictor by plotting Nitrate-N

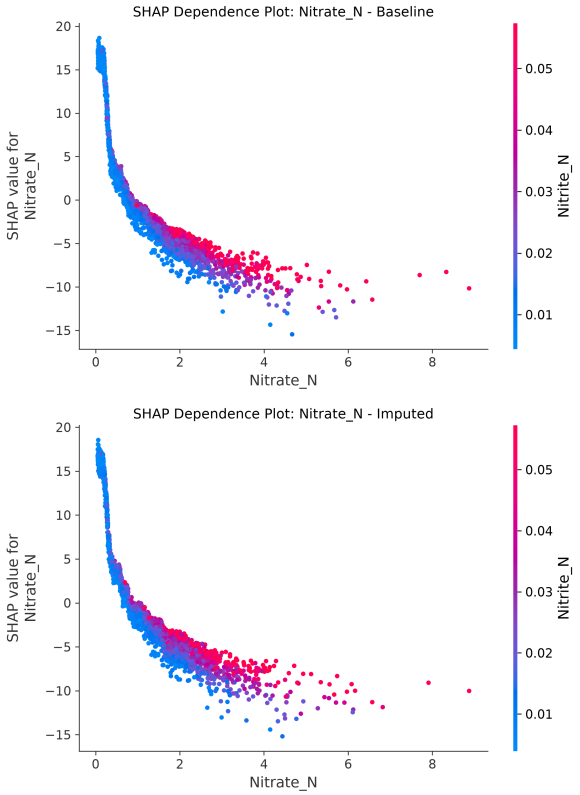


Fig. 7 Nitrate-N SHAP dependence plots for baseline (top) and imputed (bottom) test samples. The horizontal axis is the observed predictor value and the vertical axis is its contribution to the prediction. Color represents the companion predictor labeled in the original plots and does not establish a causal interaction.

values against their local SHAP contributions. Coloring denotes the companion predictor identified on each saved plot. The point distribution describes fitted-model behavior within the evaluated samples; it is not a controlled estimate of a nitrate intervention or evidence of a pollution source.

4.5.4 Local Sample-Level Decomposition

The local-analysis script selects the records with the lowest and highest *predicted* CCME-WQI within each test set. Figures 8 and 9 reproduce all four saved waterfall explanations. Each panel explains only its selected record; the examples are prediction extremes rather than a random or representative sample. The saved contribution tables identify Ammonia-N as the largest-magnitude negative contribution in both lowest-prediction examples, and Nitrate-N as the largest positive

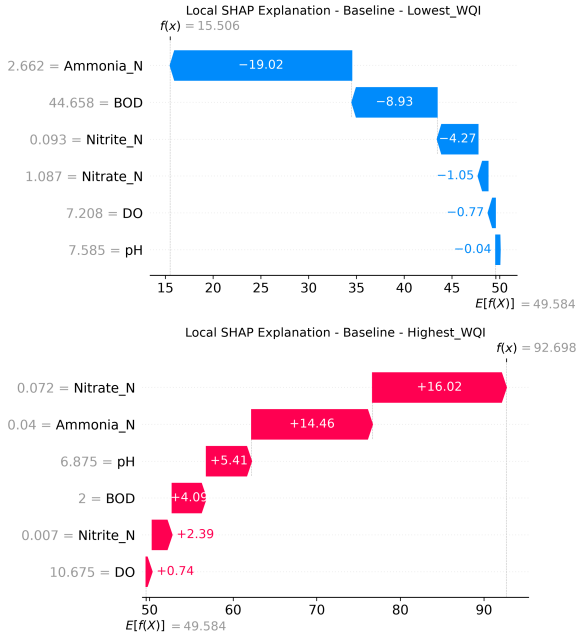


Fig. 8 Local SHAP waterfall explanations for the baseline test records with the lowest (top) and highest (bottom) predicted WQI. The six signed contributions bridge the expected model output $E[f(X)]$ and individual prediction $f(x)$.

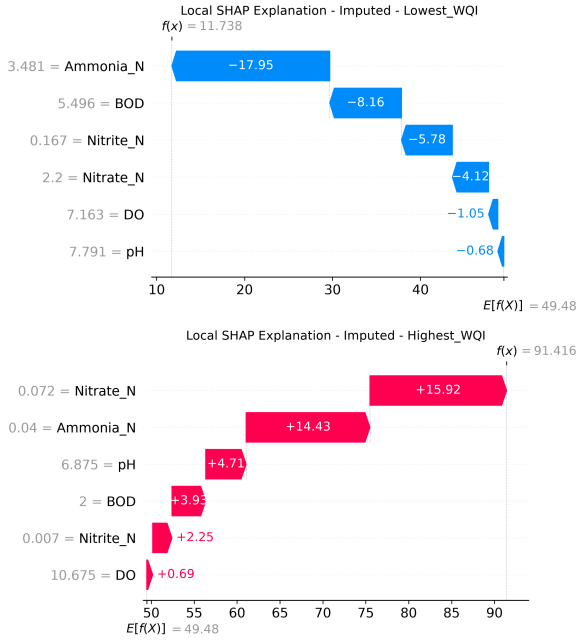


Fig. 9 Local SHAP waterfall explanations for the imputed test records with the lowest (top) and highest (bottom) predicted WQI. Each panel describes a single prediction; attribution magnitudes do not identify physical causes or verified pollution sources.

contribution in both highest-prediction examples. These local observations are distinct from the aggregate ranking.

4.6 Recommendation-System Results

For each evaluated record, the system provides a predicted WQI, assigns a corresponding water-quality category, and determines a priority based on the predicted category and the number of parameters exceeding their specified guideline objectives. SHAP-based local explanations were used to identify the water-quality parameter with the strongest negative influence on each model prediction. This parameter was then linked to predefined advisory messages describing possible contributing pressures and recommended follow-up actions.

4.7 Dashboard Validation

To verify the dashboard's functionality under degraded water-quality conditions, a stress-test case was evaluated using the input values: BOD = 8.0 mg/L, DO = 2.0 mg/L, Ammonia-N = 0.5 mg/L, Nitrite-N = 1.0 mg/L, Nitrate-N = 5.0 mg/L, and pH = 8.5. For the selected stress-test case, the dashboard predicted a CCME-WQI of approximately 19.61, placing the water quality in the Poor category. The system assigned a High monitoring priority, with all five guideline-assessed parameters exceeding their configured thresholds. It also successfully generated the local SHAP explanation along with the associated pressure and recommendation messages. However, this test represents a functional validation of the implemented workflow and should not be considered field validation or evidence of real-world environmental performance.

5 Discussion

5.1 Justification for Selecting CatBoost

Although XGBoost achieved a marginally lower Mean Absolute Error (2.9187 vs. 2.9632), the manuscript benchmark reports CatBoost with the lowest overall Root Mean Squared Error (3.9849 baseline, 4.0024 imputed) and the highest coefficient of determination (R^2 = 0.9286). RMSE weights larger prediction errors more heavily; it

does not establish detection of hazardous events. CatBoost was selected for later analysis. No comparative SHAP-timing experiment or isolated feature-interaction evaluation is supplied, so faster explanation computation and superior interaction handling are not claimed.

5.2 System Constraints and Future Extensions

1. Parameter Scope: The model operates strictly over 6 chemical parameters. In real river basins, variables such as water temperature, electrical conductivity, seasonal discharge rate, heavy metals, and upstream land use provide valuable supplementary signals.
2. Data-Quality Scope: The project processes historical monitoring observations. The experiments do not establish performance under live sensor drift or transmission loss, and no sensor-calibration evaluation is reported.
3. Target Construction and Predictor Relationship: The CCME-WQI used as the prediction target is a constructed composite index based on five of the water-quality parameters included in the predictor set. Therefore, the reported model performance should be interpreted as estimation of the constructed WQI from its underlying measurements rather than prediction of an independently measured environmental outcome. The high predictive performance should consequently not be interpreted as evidence of equivalent performance for independently observed or future water-quality outcomes.
4. Advisory Scope: The recommendation engine generates hypothesis-driven advisories based on parameter patterns. These guidelines assist human inspectors in prioritizing field visits but do not replace certified laboratory testing.

5.2.1 Future Roadmap

Geospatial Plume Tracking: Incorporating GIS coordinates into the Streamlit dashboard to visualize downstream pollutant diffusion.

Temporal Forecasting: Future work could investigate sequence models with explicitly defined future targets and chronological validation. The current implementation has neither a forecasting task nor a validated forecast horizon.

External Cross-Basin Validation: Testing model portability against river monitoring datasets from other geographic regions.

6 Conclusion

This study develops an explainable AI-based decision-support system for estimating and interpreting the constructed station-year CCME-WQI from six predictors. It does not forecast future water quality. It uses long-term Northern Ireland Environment Agency (NIEA) observations and includes preprocessing, missingness analysis, WQI construction, model comparison, row-based validation, SHAP analysis, recommendations, and dashboard development.

The CatBoost model showed strong overall predictive performance and was selected for the later stages of the system. Five-fold cross-validation was used to assess the consistency of the model across different validation folds.

SHAP was then used to explain the predictions by examining the contributions of the input parameters. By combining the predicted WQI, guideline-exceedance information, and model interpretation, we developed a recommendation engine.

The final system is implemented as an interactive Streamlit dashboard.

The main value of the proposed system is the integration of prediction, explanation, and decision support into a single workflow, rather than the provision of only a numerical WQI prediction.

Declarations

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Competing Interests. The authors declare no competing interests.

Ethics Approval and Consent to Participate. Not applicable. This study used an environmental water-quality monitoring dataset and did not involve human participants, human subjects, or animal experimentation.

Consent for Publication. Not applicable.

Data Availability. Availability of the underlying dataset is subject to the terms and conditions under which the source data were obtained.

Materials Availability. Not applicable.

Code Availability. A public repository is not specified at the time of manuscript preparation.

Author Contributions. Chintakayala Karthikeya: Conceptualization, methodology, data analysis, model development, experimentation, and manuscript preparation.

Ameer Roshan Shaik: Data preprocessing, implementation, experimentation, and manuscript review.

Shaik Abdul Samad: Machine-learning/deep-learning implementation, explainable AI analysis, dashboard development, and manuscript preparation.

Desam Pranay Kumar Reddy: Data analysis, recommendation-system development, visualization, experimentation, and manuscript review.

All authors read and approved the final manuscript.

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[13] Project implementation files supplied for manuscript verification: `05_build_station_year_WQI.py` in `02_Models/04_WQI_Calculation/`, and `05_build_station_year_WQI_imputed.py` in `02_Models/04_WQI_New/`. Unpublished Python scripts documenting the configured objectives and eligibility rules; release version and public archival identifier not supplied.